

Pranav Kaja

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EDUCATION

University of Massachusetts Amherst

Amherst, MA

Master of Science in Business Analytics, GPA: 4.00/4.00

Sep 2024 – May 2026

- Coursework: Machine Learning, Data Management, Statistics, Data Visualization, Project Management

GITAM Deemed to be University

Bachelor of Technology in Computer Science and Engineering

Aug 2019 – May 2023

SKILLS

Analytics & Business Intelligence: Power BI (Advanced DAX), Tableau, KPI Dashboarding, Data Storytelling, Segmentation, Statistical Analysis, Demand Forecasting

Programming & Data Engineering: Python (Pandas, NumPy, Scikit-learn), SQL (PostgreSQL, BigQuery, Window Functions), R, Advanced Excel (VBA, Power Query), Flask, REST APIs, Git/GitHub, ETL Pipelines, Data Warehousing

Modeling & AI: Random Forest, Gradient Boosting, XGBoost, Logistic Regression, SMOTE, LLM Integration (Gemini on Vertex AI), Feature Engineering

EXPERIENCE

UMass Amherst Dining Services

Amherst, MA

Student Supervisor

Sep 2024 – May 2026

- Cut manual entry **90%** by engineering an automated ETL pipeline in VBA and Power Query that parsed daily transaction data across 37 retail locations
- Segmented **15,000** daily patrons by volume and product mix, reallocating staffing during peak windows to raise throughput at flat labor cost
- Lifted yield **27%** by auditing data across prep and service stages to surface root-cause inefficiencies

Acmegrade

Data Science Intern

May 2022 – Jul 2022

- Lifted conversion-prediction accuracy **15%** over baseline by training Random Forest and Gradient Boosting classifiers on a 100,000-record client portfolio

Esports SIG – GITAM

Operations and Finance Lead

Sep 2022 – Jul 2023

- Tracked finances across 7 concurrent programs, tripling turnout to **1,800+** through data-led scheduling; secured **100%** external sponsor funding

Crescendo Magazine – GITAM

Marketing and Social Media Coordinator

Sep 2022 – Jul 2023

- Led a 5-person team across two campaign cycles, growing audience engagement **170%**

TECHNICAL & OPERATIONS PROJECTS EXPERIENCE

Two-Stage Fraud Detection | XGBoost, Google Gemini (LLM)

- Built a two-stage pipeline: XGBoost scores 284,807 transactions, a Gemini agent triages high-risk flags
- Lifted the model's AUC from 0.60 to **0.978** despite fraud being under 1% of data, catching 84% of it

Overheard: Social Response-Gap Engine | BigQuery, Vertex AI (Gemini), Python

- Classified **3,662** comments from 17 consumer brands' own social posts with Gemini on Vertex AI, ranking every missed customer question and complaint by response priority
- Hand-verified 34 live threads to calibrate scraper reply blindness, publishing the corrected finding that **7 in 10** directed customer messages get no public reply
- Shipped the live league-table console and a verified missed-message index used for brand outreach

End-to-End Heart Disease Prediction Engine (Capstone) | Python, Scikit-learn, Flask

- Compared five classifiers on a 13-feature clinical dataset; Random Forest led at **82%** accuracy and was deployed as a Flask web app for real-time risk scoring

Auto Insurance Churn Prediction | Python, Scikit-learn, SMOTE

- Modeled churn on **92,849** policyholders (11.5% churn); removed a data-leakage column (post-suspension date) that had inflated an earlier model
- Applied SMOTE and compared five classifiers on ROC-AUC; Gradient Boosting won at 0.70 and was threshold-tuned to prioritize recall

CPK Production Planner | Vanilla JavaScript, HTML, CSS

- Shipped a client-side tool that pivots a daily order report into per-station production sheets, replacing a **39-tab** spreadsheet across 36 locations

TapInnovations Corporate Analytics | Power BI, DAX

- Designed executive Power BI dashboards on live B2B streams, cutting weekly review time by **30%**; custom DAX measures gave leadership direct visibility into throughput